Harry Nguyen

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Education

Georgia State University

Bachelor of Science in Computer Science · GPA: 3.75/4.0 *Expected May 2027*

- **Coursework:** Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures

Work Experience

TiMoto AI **Sep 2025 – Present**

Software Engineer

- Prevented single-zone failures from taking production down, sustaining **99.9% uptime** and cutting infrastructure cost **44% ($40–60/month)**, by architecting a multi-AZ **ECS Fargate** deployment with **Terraform IaC**, CloudWatch observability, and an auto-rollback circuit breaker pattern
- Restored **100% evaluation success rate** at **sub-50ms p99** by troubleshooting a production **gRPC deadlock** – replaced inconsistent lock-acquisition ordering under concurrent calls with a single enforced sequencing rule across the inter-service layer
- Cut incident root-cause time from unknown to **under 15 minutes** by building TiMoto's monitoring stack from zero – deployed **Prometheus** and **Grafana** metrics alongside CloudWatch dashboards across the gRPC and ML serving layers
- Eliminated **OOM failures** under concurrent production load by replacing naive batching with a **vLLM/PagedAttention** inference engine, solving KV cache memory fragmentation for LLM serving
- Cut mean-time-to-identify **68%** by building a post-mortem practice with runbooks and a shared on-call rotation

Google **May 2025 – Aug 2025**

Software Engineer Intern – Chrome Browser

- Shipped a **C++ IPC transport layer** to Chrome's stable channel – serving **3B+ active users** at **sub-50ms p99** and **10K+ req/sec** – by choosing Protocol Buffers over custom serialization for schema evolution and cross-language compatibility
- Landed production-ready changes into Chrome's codebase – reviewed and adopted by senior engineers – by delivering design docs and code reviews at **95% test coverage** alongside the Chrome infrastructure team
- Turned a visibly sluggish settings search into an instant one for power users – **96% p99 latency cut** (1,200ms down to sub-50ms), zero production regressions – by profiling the hot path and replacing a mutex-contended search with a **lock-free concurrent trie**
- Ended cross-team cascading UI bugs and accelerated feature delivery **68%** across **25K+ lines** of Chromium, at **95% test coverage**, by architecting an event-driven TypeScript/React system that decoupled UI state propagation via the observer pattern

Develop for Good **May 2024 – Aug 2024**

Software Engineer Intern

- Scaled a backend **REST API** to support **500+ concurrent users** and cut deployment time **90%** by designing a stateless AWS BaaS with **JWT auth** and a GitHub Actions **CI/CD** auto-scaling policy
- Cut API response time from 3+ seconds to **sub-100ms** on 10,000+ record datasets by diagnosing an N+1 query bottleneck and redesigning the data layer with **PostgreSQL indexing**

Personal Projects

Pulumi – Contributor | *Go, TypeScript, Infrastructure-as-Code (IaC)*

- Submitted **Go** CLI features and bug fixes enabling **multi-cloud (AWS/Azure/GCP)** provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

Technical Skills

Distributed Systems: gRPC, Protocol Buffers, REST APIs, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture
Languages: Python, Go, TypeScript, C++, Java, JavaScript, SQL, Bash
Frameworks & Databases: Django, FastAPI, Node.js, PostgreSQL, Redis, MongoDB, DynamoDB
Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Kubernetes, Docker, Terraform, CI/CD, CloudWatch, Prometheus, Grafana, GCP
ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain
AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor